



**POLITECNICO
MILANO 1863**

**COMPUTER SCIENCE AND ENGINEERING
INTERNET OF THINGS (IoT)
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Challenge #3

Node-Red & LoRaWAN

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Part 2| Exercise

A LoRaWAN network in Europe operates at carrier frequency 868 MHz with bandwidth 125 kHz. The network is composed of one gateway and 40 sensor nodes. Each sensor node transmits packets with payload size 20 bytes according to a Poisson process with intensity $\lambda=2$ packets/minute.

EQ1:

Determine the largest LoRa spreading factor (SF) that guarantees a packet success rate of at least 75%. Assume that all nodes use the same SF and adopt a simplified model where packet transmissions are uncoordinated and may overlap in time. To compute the airtime of a packet, use the TTN airtime calculator: <https://www.thethingsnetwork.org/airtime-calculator>

We use the pure ALOHA model, where the success rate is given by:

$$- \text{ Succ. Rate } = \frac{S}{G} = e^{-2G} = e^{-2N\lambda t}$$

Based on the data provided:

- $N = 40$ sensor node
- $\lambda = 2$ packets/minute
- Required success rate $\geq 75\%$

We impose the condition:

$$- e^{-2N\lambda t} \geq 75\% \\ \Rightarrow t \leq -\frac{\ln(0.75)}{2 \cdot N \cdot \lambda} = -\frac{\ln(0.75)}{2 \cdot 40 \cdot \frac{2}{60}} \simeq 0.10788 \text{ s} = 107.88 \text{ ms}$$

Therefore, we estimate that the airtime must be less than 107.88 ms.

Then by using the TTN airtime calculator and by entering the required parameters:

- Input Bytes = 20 Bytes
- Region = EU868
- Bandwidth (BW) = 125 kHz

LoRaWAN airtime calculator

Don't waste your airtime. Be mindful about the spreading factors you are using and always go for the highest transmission speed possible as this leads to a longer battery life and less gateway utilization.

Input Bytes	Spreading Factor	Region	Bandwidth
20	7	EU868	125 kHz

we obtain the airtime for different spreading factor (SF):

- SF7: Time on air = 71.9 ms
- SF8: Time on air = 133.6 ms
- SF9: Time on air = 246.8 ms

We observe that by increasing the spreading factor leads to a higher airtime. Since SF8 already exceeds the maximum allowed airtime of 107.88 ms, all higher spreading factors (SF9-SF12) will also violate the constraint.

Therefore, only SF7, with an airtime of 71.9 ms, satisfies the requirement.

To verify this, we substitute the airtime values into the success rate formula:

- For SF7: $Succ. Rate = e^{-2 \cdot 40 \cdot \frac{2}{60} \cdot 0.0719} \simeq 0.8255 = 82.55\% (\geq 75\%)$
- For SF8: $Succ. Rate = e^{-2 \cdot 40 \cdot \frac{2}{60} \cdot 0.1336} \simeq 0.7003 = 70.03\% (< 75\%)$
- For SF9: $Succ. Rate = e^{-2 \cdot 40 \cdot \frac{2}{60} \cdot 0.2468} \simeq 0.5178 = 51.78\% (< 75\%)$

In conclusion, the largest LoRa spreading factor (SF) that guarantees a packet success rate of at least 75% is SF7, achieving a success rate of 82.55%.

EQ2:

After deploying the network in the field, the measured packet success rate is significantly lower than expected. Moreover, performance is not uniform across nodes: some nodes achieve relatively good delivery rates, while others experience very poor performance.

Which of the following actions is the most appropriate? Briefly justify your answer.

1. Decrease the number of nodes
2. Move the nodes closer to the gateway
3. Increase the spreading factor

The most appropriate action is to move the nodes closer to the gateway. Since some nodes perform much worse than others, the problem is likely caused by poor link quality instead of only network congestion. Nodes that are farther away may experience weaker signal strength and higher path loss, leading to low packet delivery rates. Reducing the number of nodes may lower collisions, but it does not improve the connection quality of weak nodes. Increasing the spreading factor can help reliability, but it also increases airtime and reduces network capacity. Therefore, moving the nodes closer to the gateway is the best solution.